P 1.62

(a) In spherical coordinates $\vec{\nabla}\cdot\vec{V} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\, v_\theta) +$

$$+\frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial\phi}$$

$$\vec{V} = \frac{\hat{r}}{r} \Rightarrow V_r = \frac{1}{r}$$

$$\vec{\nabla}\cdot\vec{V} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{1}{r}\right) = \frac{1}{r^2}$$

Using integral representation

$$\int_{\substack{\text{sphere}\\ \text{of radius } R}} \vec{V}\cdot d\vec{S} = \int_{r=R} \frac{\hat{r}}{r}\cdot r^2\sin\theta\, d\theta\, d\varphi\, \hat{r} = 4\pi R \xrightarrow[R\to 0]{} 0 \Rightarrow \text{no delta function.}$$

Also $\vec{\nabla}\cdot(r^n\hat{r}) = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 r^n) = \frac{1}{r^2}(n+2)\, r^{n+1} = (n+2)\, r^{n-1}$

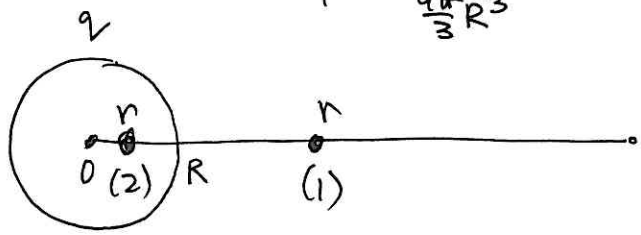
$\Rightarrow$ all cases except $n=-2$ have regular divergence.

(b) $\vec{\nabla}\times(r^n\hat{r}) = 0$ because $r^n\hat{r}$ is radial ~~fun~~ vector.

The same is seen via (1.72)

P2.21

$$\rho = \frac{q}{\frac{4\pi}{3}R^3}$$

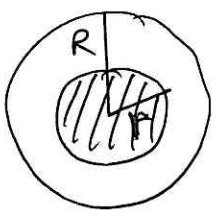


$$V(r) = -\int_{\infty}^{r} \vec{E}\cdot d\vec{l}$$

1) let $r > R \Rightarrow \vec{E}(r) = \frac{q}{4\pi\varepsilon_0 r^2}\hat{r}$ (for $r > R$)

$$V(r) = -\frac{q}{4\pi\varepsilon_0}\int_{\infty}^{r} \frac{\hat{r}'}{(r')^2}\cdot \hat{r}'dr' = \frac{q}{4\pi\varepsilon_0}\left.\frac{1}{r'}\right|_{\infty}^{r} = \frac{q}{4\pi\varepsilon_0 r}$$

2) Now $r < R$. Find $\vec{E}(r)$ inside the sphere



$$\underbrace{\int \vec{E}\cdot d\vec{S}}_{E(r)\cdot 4\pi r^2} = \frac{\rho\cdot\frac{4\pi}{3}r^3}{\varepsilon_0} = \frac{q}{\varepsilon_0}\left(\frac{r}{R}\right)^3$$

$$\Rightarrow E(r) = \frac{q}{4\pi\varepsilon_0 r^2}\cdot\left(\frac{r}{R}\right)^3 \text{ for } r < R$$

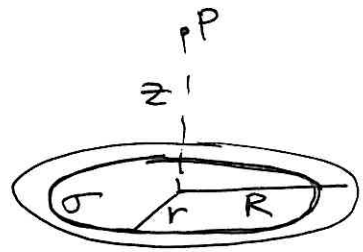
$$V(r) = -\int_{\infty}^{R} \vec{E}\cdot d\vec{l} - \int_{R}^{r} \vec{E}\cdot d\vec{l} =$$

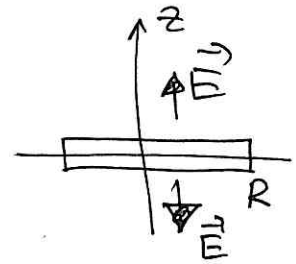
$$= \frac{q}{4\pi\varepsilon_0 R} - \int_{R}^{r} \frac{q\, r'}{4\pi\varepsilon_0 R^3}dr' =$$

$$= \frac{q}{4\pi\varepsilon_0 R} - \frac{q}{4\pi\varepsilon_0 R^3}\frac{1}{2}(r^2 - R^2) = \frac{q}{8\pi\varepsilon_0 R^3}(3R^2 - r^2).$$

P 2.25 (c)

$V(z) = ?$





$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\sigma(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\, da'$$

Consider ring of radius r and width dr.

Its area $da' = 2\pi r\, dr$

Distance to P : $\sqrt{z^2+r^2}$

$$V(z) = \frac{1}{4\pi\varepsilon_0} \int_0^R \frac{\sigma \cdot 2\pi r\, dr}{\sqrt{z^2+r^2}} = \frac{\pi\sigma}{4\pi\varepsilon_0} \int \frac{d(r^2)}{\sqrt{r^2+z^2}} =$$

$$d\left(\sqrt{r^2+z^2}\right) = \frac{1}{2}\, \frac{d(r^2)}{\sqrt{r^2+z^2}}$$

$$= \frac{2\pi\sigma}{4\pi\varepsilon_0} \int_0^R d\left(\sqrt{r^2+z^2}\right) = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+z^2} - |z|\right)$$

$$\vec{E}(\vec{r}) = -\vec{\nabla} V(\vec{r}) \underset{\text{use (1.79)}}{=} -\left(\frac{\partial V}{\partial s}\hat{s} + \frac{1}{s}\frac{\partial V}{\partial \phi}\hat{\phi} + \frac{\partial V}{\partial z}\hat{z}\right)$$

$$= -\frac{\partial V}{\partial z}\hat{z} = -\frac{\sigma}{2\varepsilon_0}\left(\frac{2z}{2\sqrt{R^2+z^2}} - \text{sign}(z)\right)\hat{z} =$$

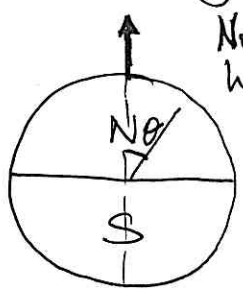
$$= \frac{\sigma}{2\varepsilon_0}\left(\text{sign}(z) - \frac{z}{\sqrt{R^2+z^2}}\right)\hat{z}.$$

Note that for $R \to \infty$ $\quad \vec{E} = \frac{\sigma}{2\varepsilon_0}\begin{cases} +\hat{z}, & \text{for } z>0 \\ -\hat{z}, & \text{for } z<0 \end{cases}$

P2.43 $\vec{F} = \int \rho(r')\, \vec{E}_s(r')\, dV'$; $\rho = \frac{Q}{\frac{4\pi}{3}R^3} = \frac{3Q}{4\pi R^3}$

North hemisphere

electric field from southern hemisphere



Since Northern hemisphere can not excert force on itself we can add $\vec{E}_N$ to the integrand:

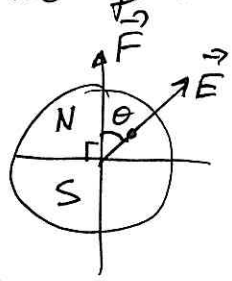
$$\vec{F} = \int_N \rho(r')\left(\vec{E}_s(r') + \vec{E}_N(r')\right) dV'$$

$$\left(\text{so that} \quad \int_N \rho(r')\, \vec{E}_N(r')\, dV' = 0\right)$$

But $\vec{E}_s(r') + \vec{E}_N(r') = \vec{E}(r')$ = electric field at point $\vec{r}'$ inside sphere.

$$\vec{E}(r') = \frac{Q\, r'}{4\pi\varepsilon_0 R^3}\, \hat{r}'$$

Clearly only z-component of $\vec{E}$ matters ⇒ take $E\cos\theta$.

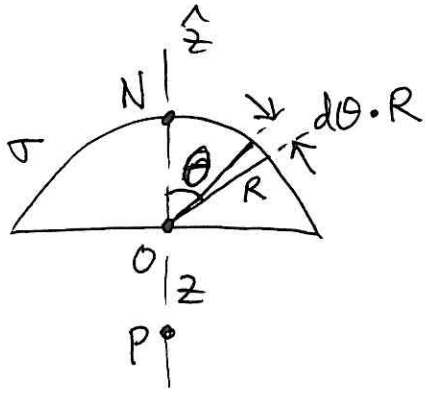


$$F = \int_N \rho \cdot \frac{Q\, r\cos\theta}{4\pi\varepsilon_0 R^3}\, 2\pi r^2 dr \sin\theta\, d\theta =$$

$$= \frac{Q\rho \cdot 2\pi}{4\pi\varepsilon_0 R^3} \underbrace{\int_0^{\pi/2} d\theta \sin\theta\cos\theta}_{\frac{1}{2}\sin^2\theta\big|_0^{\pi/2} = \frac{1}{2}} \underbrace{\int_0^R dr\, r^3}_{\frac{1}{4}R^4} = \frac{1}{4\pi\varepsilon_0}\frac{3Q^2}{16R^2}$$

$\rho = \frac{3Q}{4\pi R^3}$

P2.44



Area of charged surface element $= 2\pi R \sin\theta (R\,d\theta)$
$= 2\pi R^2 \sin\theta \, d\theta$

$(\text{Distance to } P)^2 = (z + R\cos\theta)^2 + (R\sin\theta)^2$

Consider point P on axis $\hat{z}$ of the hemisphere.

$$V(z) = \frac{\sigma}{4\pi\varepsilon_0} \int_0^{\pi/2} \frac{2\pi R^2 \sin\theta \, d\theta}{\left[z^2 + R^2\cos^2\theta + 2zR\cos\theta + R^2\sin^2\theta\right]^{1/2}}$$

$$= \frac{\sigma R^2}{2\varepsilon_0} \int_1^0 \frac{-d(\cos\theta)}{\left[z^2 + R^2 + 2zR\cos\theta\right]^{1/2}}$$

let $s = 2zR\cos\theta$

$$V(z) = \frac{\sigma R^2}{2\varepsilon_0} \frac{1}{2zR} \int_0^{2zR} \frac{ds}{\sqrt{z^2 + R^2 + s}} =$$

$$= \frac{\sigma R^2}{2\varepsilon_0} \frac{1}{2zR} \, 2\sqrt{z^2 + R^2 + s}\,\Big|_0^{2zR} = \frac{\sigma R}{2\varepsilon_0} \cdot$$

$$\cdot \; \frac{|z+R| - \sqrt{z^2 + R^2}}{z}$$

$$V(\text{point } N) = V(z = -R) = \frac{\sigma R}{2\varepsilon_0} \frac{0 - \sqrt{2R^2}}{-R} = \frac{\sigma R}{2\varepsilon_0}\sqrt{2}$$

$$V(\text{point } O) = V(z = 0) = \frac{\sigma R}{2\varepsilon_0} \frac{R + z - \sqrt{R^2 + z^2}}{z} = \frac{\sigma R}{2\varepsilon_0}.$$

Take $z \to 0$ limit by Taylor expanding: $\lim_{z \to 0} \frac{R + z - R\sqrt{1 + \frac{z^2}{R^2}}}{z} = 1$

$$\Rightarrow \; V(N) - V(O) = \frac{\sigma R}{2\varepsilon_0}\left(\sqrt{2} - 1\right)$$